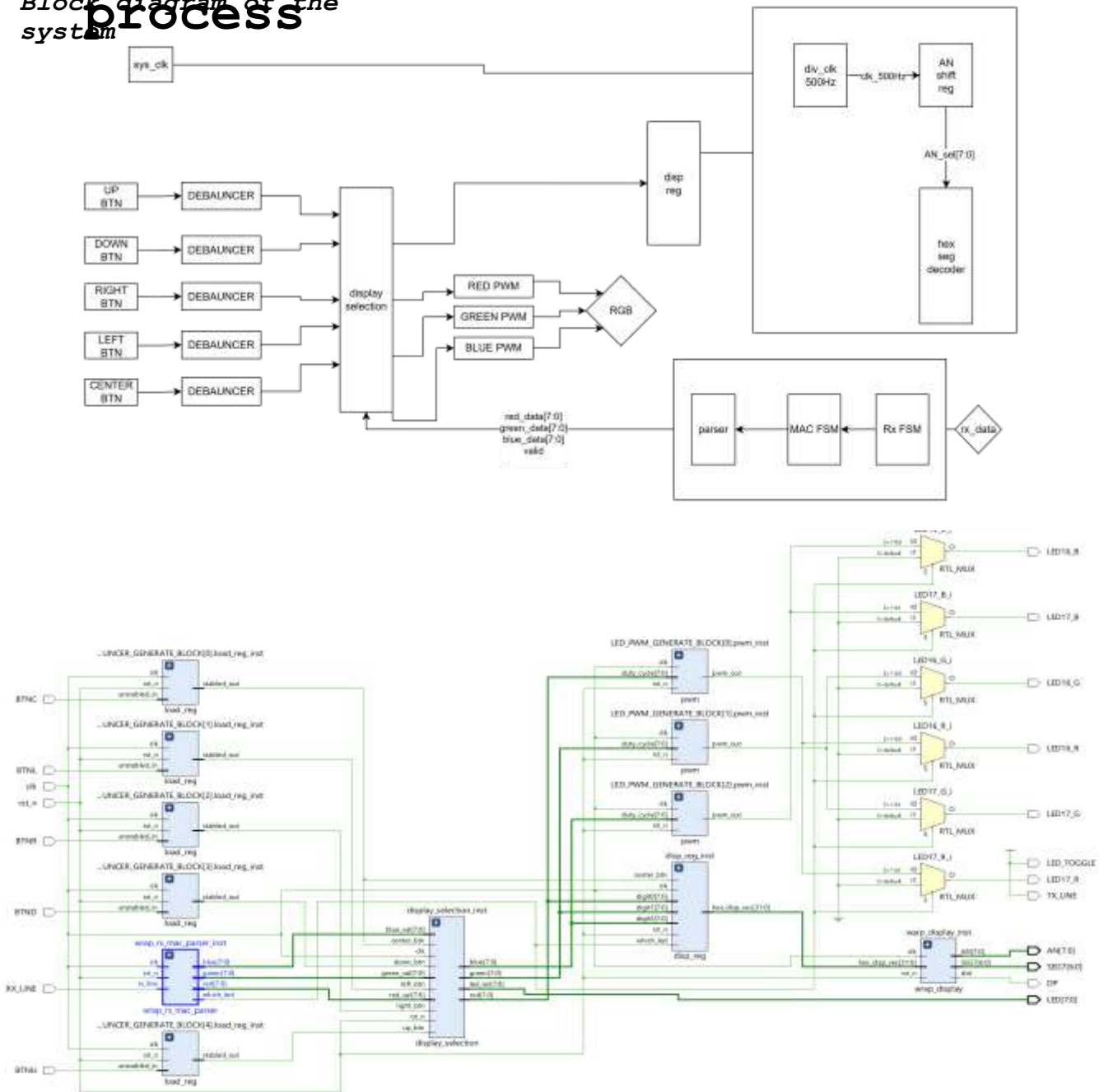


Design

Block diagram of the system process



I/O:

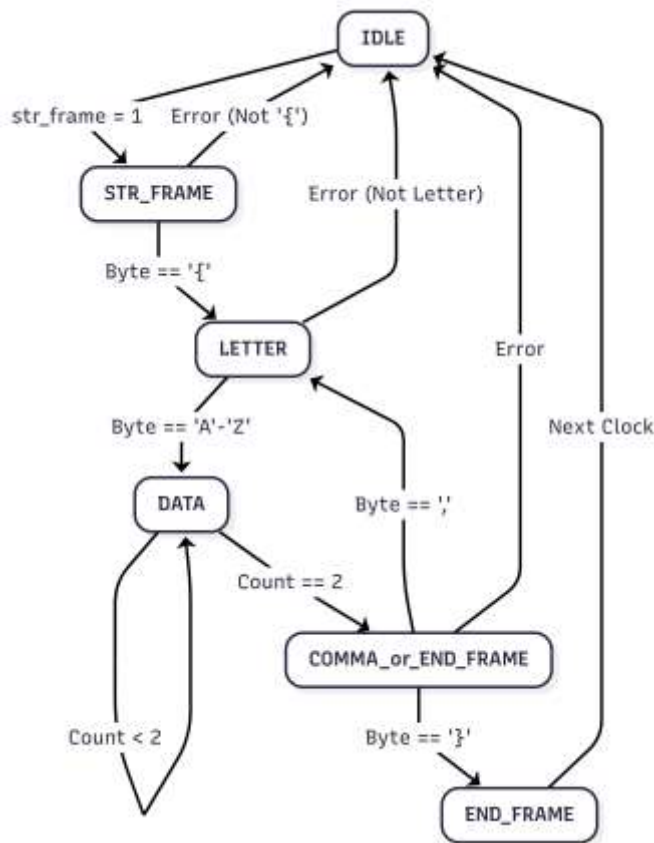
- BTNC - loading and Data.
- BTNL/BTNR - moving between config states.
- SEG7/DP - seven segment display output.
- AN - digit shift register output.
- LED - config state indicators.
- RX LINE - the Rx serial data from the computer.
- RGB - out pwm output we control with the Rx input and BTNS.

Blocks:

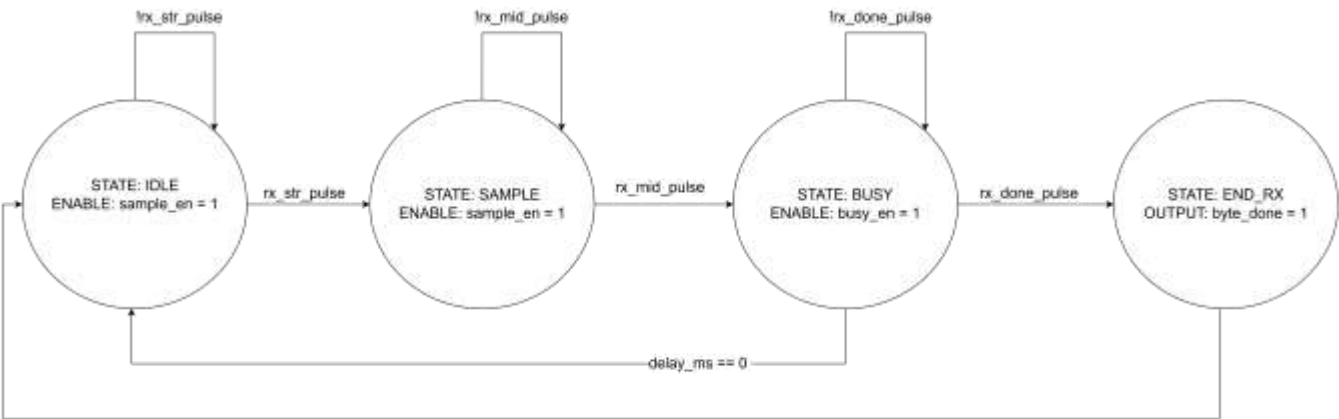
- Load_reg- debouncer for the BTNS.
- Wrap_Rx_Mac_Parser - including all the data received path components
- Disp_reg - conducting the 7 segment display value in hex for presentation as a 32bit vector.
- Wrap_display - displaying the disp register value in the 7 segment (clk div, shift reg, disp decoder)

FSM design

mac_fsm



Rx_fsm



Rx_fsm_ff

State :	New Encoding :	Previous Encoding :
IDLE :	0001 :	00
SAMPLE :	0010 :	01
BUSY :	0100 :	10
END_RX :	1000 :	11

INFO: [Synth 8-3354] encoded FSM with state register 'got_reg' using encoding 'one-hot' in module 'rx'

State :	New Encoding :	Previous Encoding :
IDLE :	000 :	000
STR_FRAME :	001 :	001
LETTER :	010 :	010
DATA :	011 :	011
COMMA_or_END_FRAME :	100 :	100
END_FRAME :	101 :	101

INFO: [Synth 8-3354] encoded FSM with state register 'got_reg' using encoding 'sequential' in module 'mac_fsm'